

Home Theatre Backlight Unit



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If you have watched your LCD/plasma television late at night in complete darkness with all lights switched off, you might have

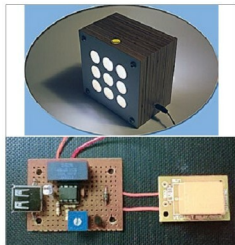


Fig. 1: Author's prototype

experienced an irritating eye strain or headache after some time. The reason behind this is that, the lone light source, the television set, has an ever-shifting level of light as the images roll across the screen, sometimes rather too dark and other times very bright. With this rapid changing of light, your pupils dilate and contract to adjust to the change. The proven way to address this is to shrink the pupils with some incidental lighting, or even better, backlighting.

Backlighting works best when a flat-panel TV is set up on a TV stand close and parallel to the wall. Backlighting offers an advantage over incidental lighting; it delineates the black bezel around the screen and highlights it. This heightens the already high contrast of the black

backlight system based on inexpensive white/blue LEDs.

The unit is compact and has a rich light output offered by a total of nine LEDs. As shown in the circuit diagram, the system is an improved version of the classic ambient-light sensor circuitry.

The circuit needs a supply voltage of 12V, but care should be taken to ensure that adequate current is available for the whole system. (In dormant state, the circuit draws a current of about 10mA and in operating state, a current determined by the electromagnetic relay and LEDs.) The backlight unit may be encased using a small homemade wooden box as shown in Fig. 1.

Circuit and working

Circuit diagram of the home theatre backlight unit is shown in Fig. 2. It is built around light-dependent resistor LDR1, op-amp LM358 (IC1), transistor BC547 (T1), nine LEDs (LED1 through LED9), 12V, 1C/O relay (RL1) and a few other components.

The LDR has very low resistance when exposed to high-intensity light and very high resistance when it is in the dark. LDR1 and resistor R1 form a voltage divider connected at the inverting (IN1-) input of IC1.

Non-inverting (IN1+) input of IC1 is connected to 10-kilo-ohm potentiometer VR1 to adjust the switching-threshold level as desired.

If ambient light level is sufficient, the circuit is in quiescent state. When the light level decreases, resistance of LDR1 increases, which causes voltage across resistor R1 to decrease,

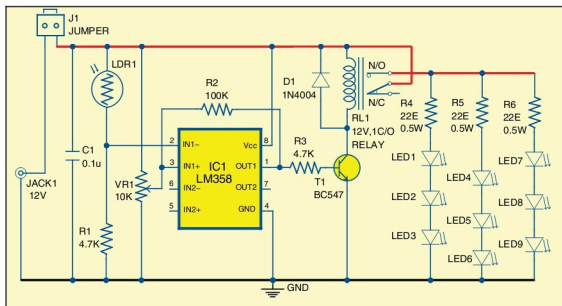


Fig. 2: Circuit diagram of the home theatre backlight unit

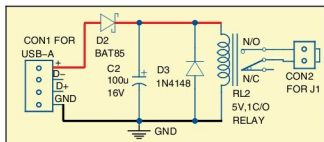


Fig. 3: Circuit diagram of the USB switch

bezel to the screen and wall, and gives it a 3D effect, which is very pleasing. Fig. 1 shows the author's prototype of the home theatre backlight unit.

Described here is a no-frills home theatre